Mental Health Analysis and Support Generation using NLP and Generative AI

Prasann Arora

M.Tech Data Science, Indian Institute of Technology Guwahati

Roll Number: 254161017 | a.prasann@iitg.ac.in

Submitted to:

Dr. Chiranjib Sur

Assistant Professor

Indian Institute of Technology Guwahati

Abstract

Mental health disorders such as depression, anxiety, and suicidal ideation have become increasingly prevalent worldwide. Individuals frequently express emotional distress through online platforms, presenting an opportunity for automated early detection and support. This paper presents an AI-based system that analyzes textual data to detect mental health indicators and generate empathetic support responses. The system integrates three large language models , Google Gemini (gemini-3.1-flash-lite-preview), Meta Llama 3.3-70B, and Alibaba Qwen3-32B , using zero-shot prompting with structured JSON output. A Retrieval-Augmented Generation (RAG) pipeline built on ChromaDB provides grounding from a curated mental health knowledge base of 515 text chunks. A LangGraphbased agentic architecture routes each query through analysis, conditional retrieval, and response generation nodes. Comparative evaluation on 400 balanced samples from the Mental Health Corpus dataset under two prompt configurations revealed meaningful precision-recall tradeoffs across models. In the baseline experiment all three models achieved near-perfect recall (0.99) due to systematic over-prediction on heavily preprocessed text. Under a refined prompt with confidence thresholding, Gemini achieved the highest accuracy (91.75%) and F1 score (91.56%), Qwen3 achieved 88.00% accuracy with F1 of 87.10%, and Llama 3 demonstrated the highest precision (96.85%) at the cost of recall (61.50%). A key finding is that aggressive NLP preprocessing significantly degrades zero-shot LLM classification by destroying the contextual signals these models depend on.

Index Terms : mental health NLP, large language models, zero-shot classification, retrieval-augmented generation, LangGraph, Gemini, Llama, Qwen3, precision-recall tradeoff, text preprocessing

I. Introduction

Mental health is a critical dimension of human wellbeing, yet access to timely support remains limited globally. The World Health Organization estimates that more than 970 million people live with a mental disorder, with treatment gaps persisting particularly in low- and middle-income settings. Digital platforms have become a primary medium through which individuals express emotional distress, presenting a rich source of signals for automated mental health monitoring.

Recent advances in Natural Language Processing (NLP) and Large Language Models (LLMs) have opened new avenues for understanding and responding to mental health text at scale. Unlike traditional machine learning classifiers that require large labeled training datasets and manual feature engineering, modern LLMs can perform zero-shot classification and generate nuanced, context-aware responses with minimal task-specific supervision.

This project develops an end-to-end AI system for mental health text analysis and support generation. The system combines structured zero-shot analysis across three LLMs, a RAG pipeline for knowledge-grounded response generation, and an agentic decision graph that orchestrates these components dynamically. This paper reports on system design, implementation, exploratory data analysis across three real-world datasets, and a comparative model evaluation including an investigation of the precision-recall tradeoff under two different prompt configurations.

II. Related Work

Early approaches to automated mental health text analysis relied on lexical features and traditional classifiers. Coppersmith et al. demonstrated that language patterns on social media can predict mental health conditions [1]. The introduction of transformer-based models, particularly BERT [2], significantly improved classification accuracy by capturing contextual semantics rather than relying on surface-level features.

More recently, LLMs have shown strong zero-shot capabilities on clinical NLP tasks. Approaches using GPT-family models for depression detection have demonstrated competitive performance against supervised baselines [3]. RetrievalAugmented Generation [4] has further enhanced generative systems by grounding responses in domain-specific knowledge, reducing hallucination in sensitive domains such as mental health support.

LangGraph [5] represents a recent development in agentic AI, enabling stateful multi-step pipelines with conditional routing. Its application to mental health support systems , where routing between crisis intervention, retrieval, and empathetic response generation is essential , has not been extensively studied, motivating the present work.

III. Datasets

Three publicly available mental health datasets from Kaggle were utilized. Table I provides a summary.

Dataset	Source	Rows	Cols	Primary Use
Mental Health in Tech Survey	OSMI/Kaggle	1,259	27	EDA , workplace patterns
Mental Health Corpus	Kaggle	27,977	2	Model evaluation benchmark
Suicide Watch	Kaggle	232,074	3	EDA , crisis language patterns

TABLE I. Dataset Summary

A. Mental Health in Tech Survey (Dataset 1)

This dataset contains 1,259 survey responses from technology industry professionals worldwide, collected by Open Sourcing Mental Illness (OSMI) in 2014. The 27 columns include demographic variables (age, gender, country), workplace characteristics (company size, remote work, self-employment), and mental health indicators (family history, treatment-seeking behavior, work interference frequency, and willingness to discuss mental health with supervisors and coworkers). This dataset was used exclusively for exploratory data analysis.

B. Mental Health Corpus (Dataset 2)

This dataset contains 27,977 Reddit posts labeled as either mental health related (label = 1, 13,838 posts) or not related (label = 0, 14,139 posts) , a near-balanced distribution. The text is heavily preprocessed: lowercased, punctuation removed, and stopwords eliminated, producing sparse keyword-style text. The dataset served as the primary benchmark for model comparison. A balanced subsample of 400 posts (200 per class) was drawn with random_state = 42 for reproducibility.

C. Suicide Watch Dataset (Dataset 3)

This large dataset contains 232,074 Reddit posts from r/SuicideWatch and r/teenagers, labeled as suicide or non-suicide with an exact 50/50 class split. Unlike Dataset 2, this dataset contains raw, unprocessed text providing full natural language context. Suicide posts average 203 words versus 61 words for non-suicide posts , a 3.3x difference. This dataset was used for exploratory data analysis only; model evaluation on this dataset is planned for future work.

IV. Exploratory Data Analysis

Exploratory data analysis was conducted across all three datasets to characterize the data landscape and motivate design decisions. Figures 1 through 3 present the key visualizations produced.

A. Mental Health in Tech Survey (Dataset 1)

Six charts were produced for Dataset 1 (Figure 1). After standardizing the gender field into Male, Female, and Other, female respondents showed a higher treatment-seeking rate relative to non-treatment compared to male respondents. The age distribution is concentrated between ages 25–40 with a mean of 32.1 years. Work interference analysis revealed 46.7% report interference "Sometimes" and 14.5% "Often" , over 60% experiencing regular work interference. The United States dominates country distribution, followed by the United Kingdom and Canada. Treatment rate by company size ranges approximately 45–58% across all sizes, with the smallest companies (1–5 employees) showing the highest rate. Family history is clearly correlated with treatment-seeking behavior.

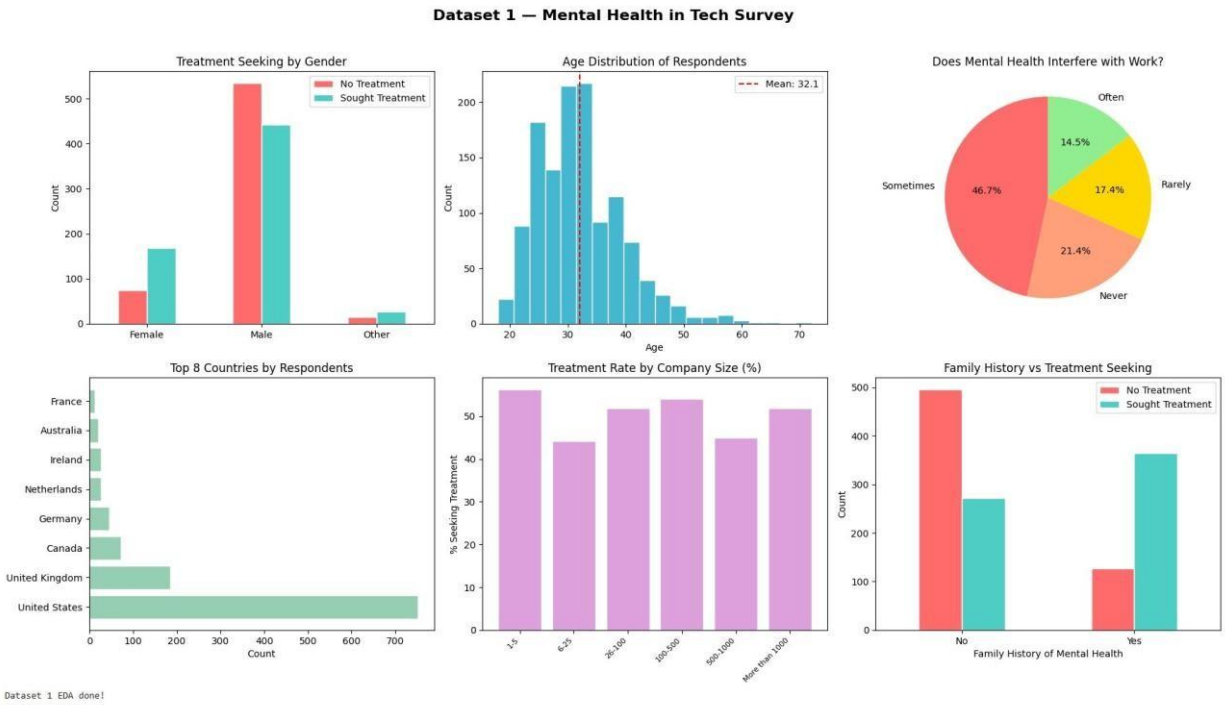

Fig. 1. EDA on Dataset 1 , Mental Health in Tech Survey: treatment by gender, age distribution, work interference, top countries, treatment rate by company size, and family history vs treatment seeking.

B. Mental Health Corpus (Dataset 2)

Three statistical charts and two word clouds were produced for Dataset 2 (Figure 2). The label distribution confirms nearbalance: 14,139 non-related posts and 13,838 mental health posts. Mental health posts average 97 words versus 47 words for non-related posts , a 2x difference. Text length distribution shows mental health posts extend to longer character counts. Word cloud analysis reveals that non-mental health posts are dominated by "film," "filler," and general narrative terms, while mental health posts cluster around "feel," "life," "know," "want," "think," and "friend." The prominence of "filler" in label = 0 posts reflects a preprocessing artifact where stopword placeholders were inserted in some entries.

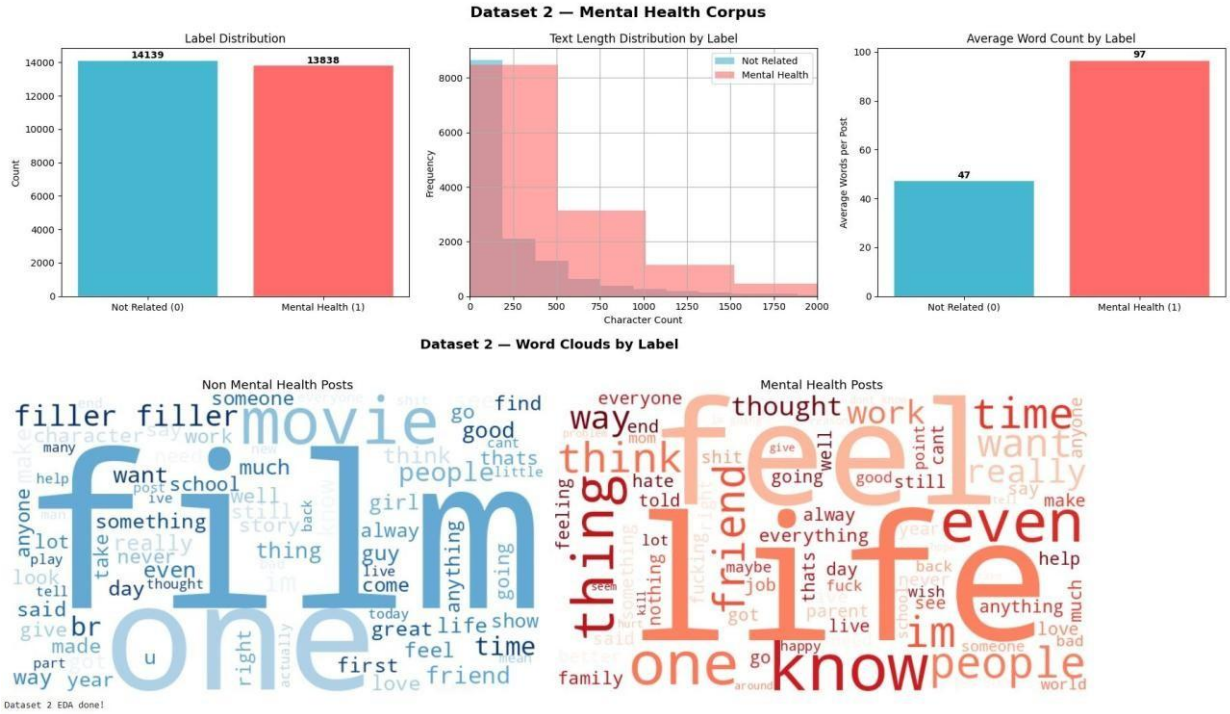

Fig. 2. EDA on Dataset 2 , Mental Health Corpus: label distribution, text length distribution, average word count by label, and word clouds for non-mental-health vs mental health posts.

C. Suicide Watch Dataset (Dataset 3)

Three statistical charts and two word clouds were produced for Dataset 3 (Figure 3). The class distribution pie chart confirms the exact 50/50 split. Text length distribution shows a clear separation with suicide posts extending significantly further , many exceeding 1,500 characters. Suicide posts average 203 words versus 61 words for non-suicide, the largest word count disparity across all three datasets. Word cloud analysis of suicide posts reveals distinctive vocabulary dominated by "want," "life," "feel," "know," and crucially the word "die," confirming crisis-specific language patterns. Non-suicide posts show general social vocabulary including "mom," "friend," "school," and "know."

Fig. 3. EDA on Dataset 3 , Suicide Watch Dataset: class distribution, text length distribution, average word count by class, and word clouds for non-suicide vs suicide posts.

V. System Architecture

The system is composed of four integrated components: (1) a zero-shot analysis module powered by LLMs, (2) a RAG knowledge retrieval pipeline, (3) a LangGraph agentic orchestration graph, and (4) a Streamlit web interface for interactive access.

A. Zero-Shot Analysis Module

The analysis module sends text to an LLM with a structured prompt requesting a JSON object with fields: primary_concern (depression, anxiety, crisis, stress, loneliness, or unclear), severity_level (low, moderate, high, critical), key_themes (list), safety_concerns (boolean), analysis_summary (two sentences), and support_response (empathetic reply). Three models were integrated: Google Gemini accessed via the google-genai Python library, and Meta Llama 3.370B and Alibaba Qwen3-32B accessed via the Groq inference API. Mixtral-8x7B was initially intended as the third model but was decommissioned by Groq mid-project; Qwen3-32B was substituted.

B. RAG Knowledge Pipeline

Five expert-curated mental health documents were compiled covering: depression (symptoms, treatment), anxiety disorders (types, CBT therapy), crisis intervention (988 Lifeline, warning signs), coping strategies (ten APA-based techniques), and workplace mental health. Each document was split into overlapping chunks of 500 characters with 100character overlap, yielding 515 total chunks. Each chunk was embedded using gemini-embedding-001 and stored in ChromaDB. At inference time, semantic similarity search retrieves top-k relevant chunks to ground the response.

C. LangGraph Agentic Architecture

The system employs a LangGraph StateGraph with three nodes and one conditional routing decision. The state schema carries: user_input, analysis_results, retrieved_context, and final_response.

Node	Function	Activation Condition
analyze_input	Zero-shot LLM analysis	Always , entry point
retrieve_info	ChromaDB semantic search	If any concern detected OR safety_concerns = True
generate_response	Empathetic reply generation	After retrieval, or directly if no concern

TABLE II. LangGraph Node Definitions

The route_decider evaluates the analysis output: if safety_concerns is True, severity_level is moderate/high/critical, or any key_themes are detected, the graph routes to retrieve_info before generating a response. This ensures at-risk users always receive knowledge-grounded responses rather than parametric-only generation.

D. Web Interface

A Streamlit application provides: a free-form text input area, three sample buttons (Depression, Anxiety, Stress) using Streamlit session_state for correct re-rendering, an Analyze button invoking the full pipeline, and structured output showing primary concern, severity metric, key themes, safety alert with 988 crisis line reference, analysis summary, and support response. The application is deployed on Streamlit Community Cloud.

VI. Model Comparison Study

A. Experimental Setup

The comparative evaluation used Dataset 2 (Mental Health Corpus) with a balanced sample of 400 posts , 200 labeled as mental health related (label = 1) and 200 labeled as non-related (label = 0), drawn using stratified sampling with random_state = 42 for reproducibility. All three models received identical prompts for each sample. The is_mental_health_related boolean field in each model's JSON output was used to generate binary predictions. Performance was measured using accuracy, precision, recall, and F1-score via scikit-learn.

B. Experiment 1 : Baseline Prompt

The baseline prompt requested a JSON object with an is_mental_health_related boolean field based on a general description of mental health indicators, without strictness constraints or confidence scoring. All three models ran successfully on all 400 samples. Table III presents the Experiment 1 results.

Model	Accuracy	Precision	Recall	F1	Predicted Positives
Gemini (gemini-3.1-flash-lite)	0.877	0.808	0.990	0.890	245 / 400
Llama 3.3-70B	0.787	0.705	0.990	0.823	281 / 400
Qwen3-32B	0.823	0.742	0.990	0.848	267 / 400

TABLE III. Experiment 1 , Baseline Prompt Results (n=400)

All three models achieved near-identical recall of 0.99, indicating they flagged almost every sample as mental health related. Gemini achieved the highest accuracy (87.7%) and F1 (89.0%), predicting 245 positives. Qwen3 achieved 82.3% accuracy with 267 predicted positives. Llama 3 showed the lowest accuracy (78.7%) with 281 predicted positives , the highest over-prediction among the three. While recall is uniformly high, precision varies considerably: Gemini (80.8%), Qwen3 (74.2%), and Llama (70.5%), indicating Gemini was the most precise despite all three over-predicting. Prediction distribution analysis confirmed the over-prediction pattern: true labels contained 200 positives and 200 negatives, while Gemini predicted 245, Llama 281, and Qwen3 267 positives.

C. Root Cause Analysis : Effect of Text Preprocessing

Investigation of Dataset 2's label = 0 samples revealed why all models over-predicted: aggressive preprocessing had destroyed natural language context. Sample non-mental-health posts after preprocessing appeared as: "okay single really meant find find soulmates high school itself dont rush," "merry christmas merry christmas merry christmas," and "get got person reading click win new car." Without stopwords, articles, and sentence structure, these posts lacked the contextual

markers distinguishing them from mental health content. LLMs rely on natural language patterns for zero-shot reasoning and defaulted to classifying ambiguous preprocessed text as potentially mental health related.

This finding represents a key methodological contribution: zero-shot LLM classifiers are sensitive to preprocessing quality. Aggressive NLP preprocessing appropriate for traditional statistical models (TF-IDF, bag-of-words) significantly degrades LLM performance by removing the contextual signals on which these models depend.

D. Experiment 2 : Refined Prompt with Confidence Thresholding

To address the over-prediction problem identified in Experiment 1, the prompt was redesigned with explicit strictness instructions and a confidence scoring mechanism. Key additions included: an instruction to flag is_mental_health_related as true only if the text "CLEARLY and EXPLICITLY" mentions emotional distress; an instruction not to over-diagnose normal conversation or mild negativity; a confidence field (0.0–1.0) reflecting certainty; and a threshold rule to set is_mental_health_related true only if confidence exceeds 0.6. At prediction time, a threshold function classified a sample as positive only if is_mental_health_related was True AND confidence >= 0.65. Table IV presents the Experiment 2 results.

Model	Accuracy	Precision	Recall	F1	Predicted Positives
Gemini (gemini-3.1-flash-lite)	0.9175	0.9372	0.8950	0.9156	191 / 400
Llama 3.3-70B	0.7975	0.9685	0.6150	0.7523	127 / 400
Qwen3-32B	0.8800	0.9419	0.8100	0.8710	172 / 400

TABLE IV. Experiment 2 , Refined Prompt Results (n=400)

The refined prompt produced substantially improved and well-differentiated results. Gemini achieved the highest accuracy (91.75%) and F1 score (91.56%), with balanced precision (93.72%) and recall (89.50%), predicting 191 positives against 200 true , near-perfect calibration. Qwen3 achieved the second-highest accuracy (88.00%) with high precision (94.19%) and recall of 81.00%, predicting 172 positives. Llama 3 showed the highest precision of all models (96.85%) but the lowest recall (61.50%), predicting only 127 positives. This indicates the confidence threshold of 0.65 was overly conservative for Llama, causing substantial under-prediction of the positive class. The model's output in the notebook is labeled "Mistral" due to the Python alias analyze_mistral = analyze_qwen applied for code compatibility.

VII. Results and Discussion

A. Model Performance Summary

Across both experiments Gemini consistently demonstrated the most balanced and reliable performance. In Experiment 1, Gemini achieved the highest accuracy (87.7%) and precision (80.8%) among the three models despite all sharing similar recall. In Experiment 2, Gemini maintained leadership with 91.75% accuracy and 91.56% F1. Qwen3-32B emerged as a strong open-weight alternative in both experiments, achieving 82.3% accuracy in Experiment 1 and 88.0% in Experiment 2. Llama 3.3-70B showed the most over-prediction in Experiment 1 (281 positives) and the most underprediction in Experiment 2 (127 positives), indicating high sensitivity to prompt strictness.

B. Precision-Recall Tradeoff Analysis

The two experiments together illustrate the precision-recall tradeoff in zero-shot LLM classification. Experiment 1 with a permissive prompt produced uniformly high recall (0.99) but poor-to-moderate precision across all models, driven by systematic over-prediction on ambiguous preprocessed text. Experiment 2 with a confidence-gated prompt substantially improved precision for all models but introduced under-prediction in Llama. For mental health monitoring applications, recall is typically prioritized , missing a genuine distress signal is more harmful than a false alarm. A uniform confidence threshold of 0.65 was overly conservative for Llama; per-model threshold calibration is recommended in future work.

C. Impact of Text Preprocessing on LLM Classification

This study provides evidence that aggressive text preprocessing , specifically stopword removal and punctuation stripping , is incompatible with zero-shot LLM classification. Traditional preprocessing was designed for statistical models where vocabulary frequency drives prediction. LLMs rely on syntactic structure, function words, and contextual coherence to infer meaning. When this structure is destroyed, LLMs cannot distinguish topic-ambiguous tokens from genuine distress

indicators, leading to the uniform over-prediction observed in Experiment 1. Raw or minimally preprocessed text should be preferred when applying LLMs to zero-shot text classification tasks.

D. Qualitative Evaluation of Support Responses

The system was evaluated qualitatively on support response generation through a four-turn simulated conversation on presentation anxiety. The LangGraph pipeline produced contextually appropriate, empathetic, and progressive responses: it acknowledged anxiety without pathologizing it; provided concrete grounding techniques (5-4-3-2-1, box breathing, physical anchoring); addressed cognitive distortions with reframing; and maintained a supportive conversational tone throughout. An additional test on a panic attack scenario confirmed that the RAG retrieval mechanism surfaced relevant CBT information which was naturally integrated into the generated response.

VIII. Implementation Details

A. Technology Stack

Component	Technology	Version / Model
LLM , Primary	google-genai Python library	gemini-3.1-flash-lite-preview
LLM , Secondary	Groq API	llama-3.3-70b-versatile
LLM , Third	Groq API	qwen/qwen3-32b
Embeddings	Google Gemini Embeddings API	gemini-embedding-001
Vector Store	ChromaDB	0.6.3 (in-memory)
Agent Framework	LangGraph	StateGraph with conditional edges
Web Interface	Streamlit	Community Cloud deployment
Evaluation	scikit-learn	accuracy, precision, recall, F1
Visualization	matplotlib, seaborn, wordcloud	EDA charts and word clouds
Runtime	Google Colab	T4 GPU, Python 3.12

TABLE V. Technology Stack

B. Key Engineering Challenges

Library Migration: The google.generativeai library was deprecated during the project. The system was migrated to google.genai using a Client-based architecture (client.models.generate_content()) replacing the legacy genai.configure() and GenerativeModel() pattern. All chat session management and response parsing were reimplemented accordingly.

Model Decommissioning: Mixtral-8x7B-32768 on Groq was decommissioned mid-project. Qwen3-32B was selected as a replacement. Its tendency to include chain-of-thought reasoning within XML tags required an additional postprocessing step to strip these tokens before JSON parsing: raw = raw.split("")[-1].strip().

API Rate Limiting: Free-tier API quotas required careful management. A 0.7-second sleep between embedding requests kept usage below the 100 RPM limit for gemini-embedding-001. Retry logic was implemented for generate_content calls. Evaluation results were persisted to CSV immediately after each run to prevent data loss from Google Colab runtime disconnections.

JSON Parsing Robustness: LLMs occasionally returned markdown-wrapped JSON or preamble text.
Post-processing strips ```json fences before parsing. For Llama, a prompt tightening step was required to prevent the model returning option lists (e.g., "depression or anxiety or crisis") instead of a single selected value. The instruction "choose EXACTLY ONE of:" resolved this reliably.

IX. Limitations

Several limitations should be acknowledged. First, model evaluation was conducted exclusively on Dataset 2 which contains heavily preprocessed text. Evaluation on Dataset 3 (Suicide Watch, raw text) would provide a more complete picture of model performance on natural language input and is planned as future work. Second, the evaluation sample size of 400 posts, while statistically adequate for comparison, may not generalize across all demographic and cultural contexts in the full dataset of 27,977 posts.

Third, the system relies entirely on zero-shot prompting without fine-tuning on mental health data. Fine-tuned models may outperform zero-shot approaches on nuanced classification. Fourth, qualitative evaluation of response quality was limited to informal inspection; a formal rubric-based or LLM-as-judge evaluation was not completed within the project scope. Fifth, the system is explicitly not a clinical tool and must not be used as a substitute for professional mental health assessment.

X. Conclusion

This paper presented an end-to-end AI system for mental health text analysis and empathetic support generation, integrating three large language models within a RAG-augmented LangGraph agentic pipeline. Exploratory data analysis across three real-world mental health datasets revealed meaningful patterns: a 2x word count difference in mental health versus non-mental-health posts, and a 3.3x word count difference in suicide versus non-suicide posts.

A two-experiment comparative evaluation produced actionable findings. In the baseline experiment, all three models achieved near-perfect recall (0.99) but with varying precision: Gemini (80.8%), Qwen3 (74.2%), and Llama (70.5%). This uniform over-prediction was traced to the aggressive preprocessing of Dataset 2 which destroyed natural language context. Under the refined confidence-gated prompt, Gemini demonstrated the best overall performance with 91.75% accuracy and 91.56% F1. Qwen3-32B was a strong open-weight alternative at 88.00% accuracy. Llama 3.3-70B achieved the highest precision (96.85%) but insufficient recall (61.50%), indicating sensitivity to confidence thresholding.

The study's primary methodological contribution is demonstrating that aggressive NLP text preprocessing is incompatible with zero-shot LLM classification , a finding with practical implications for researchers applying modern LLMs to preprocessed NLP datasets. Future work will extend evaluation to the Suicide Watch dataset for crisis detection, implement per-model threshold calibration, conduct formal LLM-as-judge response quality evaluation, and explore fine-tuning of smaller models on domain-specific mental health data.

References

[1] G. Coppersmith, M. Dredze, and C. Harman, "Quantifying mental health signals in Twitter," in Proc. ACL Workshop onComputational Linguistics and Clinical Psychology, 2014.

[2] J. Devlin, M. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for languageunderstanding," in Proc. NAACL, 2019, pp. 4171–4186.

[3] K. Lamichhane, "Evaluation of ChatGPT for NLP-based mental health applications," arXiv preprint arXiv:2304.11540, 2023.

[4] P. Lewis et al., "Retrieval-augmented generation for knowledge-intensive NLP tasks," Advances in Neural InformationProcessing Systems, vol. 33, 2020, pp. 9459–9474.

[5] LangChain, "LangGraph: Building stateful, multi-actor applications with LLMs," GitHub Repository, 2024. Available: https://github.com/langchain-ai/langgraph.

[6] T. Mikolov, K. Chen, G. Corrado, and J. Dean, "Efficient estimation of word representations in vector space," in Proc. ICLR,2013.

[7] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," Journal of Machine Learning Research, vol. 12, pp. 2825–2830,2011.

[8] Kaggle, "Suicide Detection Dataset." Available: https://www.kaggle.com/nikhileswarkomati/suicide-watch, 2021.

[9] Kaggle, "Mental Health Corpus." Available: https://www.kaggle.com/reihanenamdari/mental-health-corpus, 2021.

[10] Kaggle / OSMI, "Mental Health in Tech Survey." Available: https://www.kaggle.com/osmi/mental-health-in-tech-survey, 2014.

[11] Google DeepMind, "Gemini: A Family of Highly Capable Multimodal Models," Technical Report, 2023.

[12] Meta AI, "Llama 3: Open Foundation Language Models," Technical Report, 2024.

[13] Alibaba Cloud, "Qwen3 Technical Report," arXiv preprint, 2025.

[14] Chroma, "ChromaDB: The AI-native open-source embedding database," GitHub Repository, 2023. Available: https://github.com/chroma-core/chroma.